



Four cases of fatal acute arsenic poisoning: histopathology, toxicology, and new trends

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Abstract

Arsenic is a valuable component in tumor treatment and traditional Chinese medicine and has seen widespread use in processing, manufacturing, and agriculture. Although rare, arsenic poisoning can occur in forensic practice. Elusive pathological changes, as well as obscure clinical signs, may cause arsenic poisoning to go unrecognized. Here, we report four cases of fatal acute arsenic poisoning, with careful observation of pathological changes and collection of postmortem specimens for arsenic concentration analysis. Additionally, we reviewed six cases of fatal arsenic poisoning in the past 20 years. In the present study, microvesicular steatosis in the peripheral areas of the hepatic lobules and acute splenitis were observed, which are rare findings in acute arsenic poisoning. This study summarizes the histopathological features of arsenic poisoning and presents data on arsenic distribution. Arsenic concentrations in the liver and kidneys can increase the reliability of identifying arsenic poisoning. Furthermore, in traditional Chinese medicine-related deaths, arsenic poisoning needs more attention.

Keywords Forensic pathology · Forensic toxicology · Arsenic poisoning · Concentration distribution

Introduction

Arsenic (As) is a metalloid element that ranks 20th in abundance in the Earth's crust [1]. As occurs naturally as inorganic arsenic (mainly sulfides and oxides) or organic arsenic [2], and the former has played an essential role in processing, manufacturing, agriculture, and medicine [3, 4]. Although the use of arsenic has been replaced by safer synthetic dyes and pesticides [3], it remains readily available in remote areas [5]. Additionally, As is present in many

traditional Chinese medicines (TCM) [6], and arsenic trioxide is currently used for the treatment of acute promyelocytic leukemia (APL) [4].

Known as the “king of poisons,” As was widely used in murders centuries ago [7]. While fatal acute arsenic poisoning is not as common as it once was, it still accounts for many poisoning deaths [8]. Arsenic poisoning is primarily associated with ingestion, inhalation, and dermal absorption [9], but a case of suicide by intravenous injection of As has also been reported [10]. Currently, arsenic poisoning cases are identified by detecting arsenic in the blood. However, its elusive pathological changes and obscure clinical signs

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can make identifying arsenic poisoning difficult, and thus, it could be potentially overlooked. In such cases, arsenic concentrations in major organs and common body fluids other than the blood have been reported less frequently.

Here, we report four fatal acute arsenic poisoning cases and carefully examine the pathological changes and concentrations of arsenic in postmortem specimens. To better understand the toxic mechanisms underlying such cases, we reviewed the pathological changes and arsenic concentrations of postmortem specimens from fatal arsenic poisoning cases during the past 20 years to provide references.

Case reports

Four fatal acute arsenic poisoning victims were enrolled from Tongji Medical College, HUST. The basic information about the four victims is listed in Table 1.

Autopsy findings and histopathological examinations

A semiquantitative score was devised to evaluate histopathological changes in major organs. The scores, as described by Bedossa [11] and Xu et al. [12], were as follows: -, no lesions (0–5%); +, mild lesions (5–25%); ++, moderate lesions (25–50%); and + + +, severe lesions ($\geq 50\%$). All tissue sections were scored at the same magnification (40 $\times$, 100 $\times$, 200 $\times$, and 400 $\times$) by two investigators who were blinded to the study. The histopathological findings of the major organs are listed in Table 2.

Liver

Hepatocytes showed swelling, steatosis, cholestasis, and patchy necrosis in all cases. In cases 3 and 4, numerous small cytoplasmic vesicles of hepatocytes in the peripheral areas of the hepatic lobules were identified as lipid droplets by ORO staining, indicating the development of microvesicular steatosis (Fig. 1a, b). In the portal area, mononuclear and lymphocytic infiltrates were observed.

Kidney

Swelling and necrosis of epithelial cells were found in both the proximal and distal renal tubules, with greater severity in the proximal tubules. Renal tubular epithelial cells showed increased cytoplasmic eosinophilia and loss of nuclei. Diffuse epithelial cell fragmentation, brush border detachment, luminal dilatation, and even bare basement membrane formation were observed in some renal tubules. Protein casts were observed in tubular cavities (Fig. 1c, d). In addition, renal pelvic hemorrhage was observed in case 2.

Heart

Myocardial swelling and constriction band necrosis were the typical manifestations of the four cases (Fig. 1e), and inflammatory infiltration of interstitial tissues was found in case 4. Subendocardial (case 1) and epicardial (cases 3 and 4) hemorrhages were also noted.

Lung

Pulmonary edema and emphysema were standard features of the four cases. Cases 1 and 2 presented severe pulmonary edema, while cases 3 and 4 were more pronounced with emphysema. Alveolar hemorrhage was noted in case 3.

Brain

Cerebral edema, neuronal degeneration, and necrosis were observed. Bilateral cerebral herniation of the cerebellar tonsils was present in cases 1 and 2.

Spleen

Neutrophil infiltration of the splenic corpuscles was observed in case 2, suggesting acute splenitis (Fig. 1f).

Skin

In case 2, there was a prolapsed hemorrhoid at the anus, and a hematoma with severe cellulitis was seen upon incision. In other cases, no significant skin changes were noted.

Stomach

Hemorrhage, epithelial cell necrosis, and inflammatory infiltration were noted in the gastric mucosa of case 1.

Furthermore, congestion and large numbers of neutrophils in the blood vessels were found in many organs.

Toxicological analysis

Autopsy specimens, including cardiac blood, the liver, the kidney, stomach contents, the lung, the brain, the heart, and the spleen in cases 1 and 2 (in addition to muscle and thymus in case 1, bile, vitreous humor, and peripheral blood in case 2) as well as cardiac blood, the liver, and stomach contents in cases 3 and 4, were collected in polypropylene cryovials and stored at $-80\text{ }^{\circ}\text{C}$ until analysis. The concentration of total arsenic was determined using

Table 1 Basic information of the four cases in this study

Case No	Age/gender	Arsenic intake pathways	Manner of death	Survival time	Clinical features		Brief presentation
					Symptoms	Laboratory tests	
1	47/female	Ingestion of medication	Medical accident	9 h	Vomiting, diarrhea, abdominal pain, numbness of limbs	Neutrophilia, liver, and kidney dysfunction, myocardial injury, hypokalemia	The woman ingested a decoction of an arsenic-containing Chinese herb prescribed by a local traditional Chinese medicine practitioner to treat swollen lymph nodes in her left groin. Within 30 min, she experienced severe gastrointestinal symptoms and was immediately transferred to the hospital for prompt medical attention. Despite suspecting herbal poisoning, the exact type of poison was unclear, and no specific treatment was administered. Unfortunately, after 8 h of hospitalization, her clinical condition deteriorated and she died
2	49/female	Absorption through the anal skin	Medical accident	8 days	Localized pain, Vomiting	Mild liver dysfunction	The woman, who had mixed cricoid hemorrhoids, applied a folk prescription powder containing arsenic for topical application on prolapsed hemorrhoids. She applied the powder a total of five times, on days 0, 1, 2, 3, and 7. After the fourth application, she experienced localized pain and vomiting, prompting her to seek medical attention at the hospital. Upon examination, only mild liver dysfunction was found, and no treatment was administered. However, she died suddenly approximately 30 h after the fifth application

Table 1 (continued)

Case No	Age/gender	Arsenic intake pathways	Manner of death	Survival time	Clinical features		Brief presentation
					Symptoms	Laboratory tests	
3	69 /male	Ingestion of food	Homicide	2 days	Vomiting, diarrhea, abdominal pain, oliguria	Neutrophilia, liver, and kidney dysfunction, coagulation disorder, myocardial injury, hypokalemia, hypocalcemia	Cases 3 and 4 were a couple who frequently collected food from the roadside. A man, who had conflicts with the couple, put arsenic-containing pesticides into the ingredients of the porridge and left it at the couple's doorstep. After consuming the porridge made from the contaminated ingredients, the couple experienced vomiting and diarrhea and were transported to the hospital. Initially, the emergency physician diagnosed their symptoms as food poisoning. However, one day after admission, they were suspected of arsenic poisoning and were administered intravenous sodium dimercaptosulphonate as treatment. Despite receiving treatment, the couple's clinical condition continued to worsen, and they died 2 days after being admitted to the hospital
4	70 /female	Ingestion of food	Homicide	2 days	Vomiting, diarrhea, abdominal pain, oliguria	Neutrophilia, liver and kidney dysfunction, coagulation disorder, myocardial injury, hypocalcemia	

Table 2 Histopathological findings in the present study and previous literature on fatal arsenic poisoning

Histopathological findings	Present study				Previous literature*
	Case 1	Case 2	Case 3	Case 4	
<i>Liver</i>					
1) Edema	+	+	+	+	Zheng et al. [13]
2) Hepatocyte necrosis	+	+	+	+	Zheng et al. [13]
3) Steatosis	+	+	+	+	Lu et al. [5], Zheng et al. [13], Lech and Trela [15]
4) Microvesicular steatosis	-	-	+	+	/
<i>Kidney</i>					
1) Degeneration and necrosis of tubular epithelium	+	+	+	+	Lu et al. [5], Zheng et al. [13], Parent et al. [16]
2) Protein casts	+	+	+	+	Zheng et al. [13]
<i>Heart</i>					
1) Edema	+	+	+	+	Lu et al. [5], Lech and Trela [15]
2) Contraction band necrosis	+	+	+	+	Duncan et al. [14]
3) Hemorrhage	+	-	+	+	Lech and Trela [15]
4) Inflammation infiltration	-	-	-	+	Tournel et al. [10]
<i>Lung</i>					
1) Edema	+	+	+	+	Zheng et al. [13], Lech and Trela [15]
2) Emphysema	+	+	+	+	Lech and Trela [15]
3) Hemorrhage	-	-	+	-	Zheng et al. [13]
Cerebral edema	+	+	+	+	Lu et al. [5], Zheng et al. [13]
Splenic neutrophil infiltration	-	+	-	-	/
Multi-organ congestion	+	+	+	+	Lu et al. [5], Tournel et al. [10], Zheng et al. [13], Duncan et al. [14], Lech and Trela [15], Parent et al. [16]

* “/” means it has not been reported in literature

atomic fluorescence in case 1 and ICP–MS in cases 2, 3, and 4. The results of the toxicological analysis are presented in Table 3.

In the above cases, the source of the poisoning was clear. Based on the autopsy, histopathological examination, and toxicological analysis results, acute arsenic poisoning was the cause of death of the deceased.

Discussion

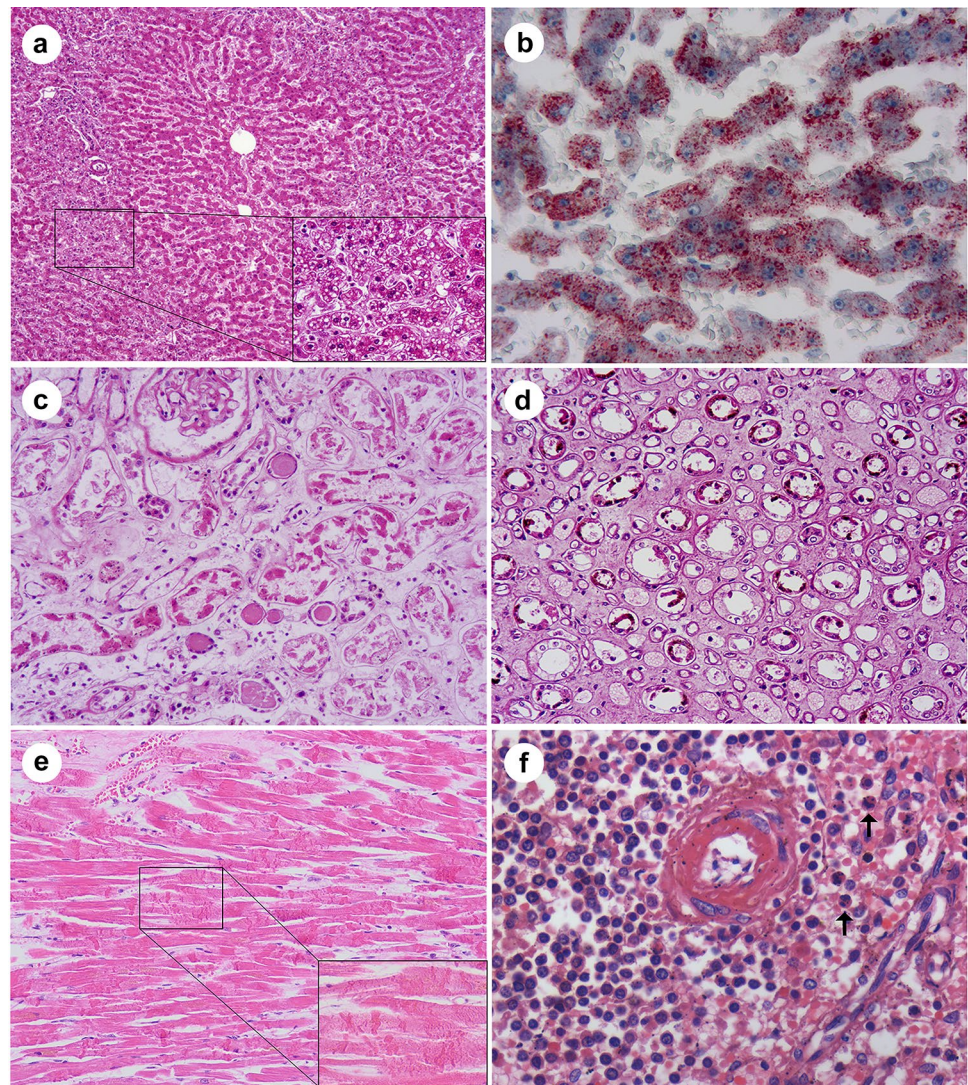
To collect pathological changes and the distribution of arsenic in arsenic poisoning deaths, we searched “arsenic poisoning,” “case report,” “fatal,” “postmortem examination,” and “autopsy” in PubMed during the last 20 years, from 2003 to 2022. We excluded papers without autopsy and arsenic concentrations and identified six fatal reports [5, 10, 13–16]. Pathological changes and the concentrations of arsenic in postmortem specimens from this study and the literature are shown in Tables 2 and 3.

Arsenic-containing minerals, such as realgar (As_4S_4), are therapeutic ingredients in many TCM prescriptions, and arsenic can also be detected in arsenic-free TCM due

to environmental pollution or adulteration [6]. While the content of arsenic in various types of traditional Chinese medicines is subject to precise requirements outlined in the Pharmacopoeia of the People’s Republic of China [17], some TCMs sold have been found to contain a higher concentration of arsenic than the permissible level in the pharmacopoeia [6]. Additionally, some TCM practitioners lack standardized professional training. These factors make it possible for patients to suffer from acute or chronic arsenic poisoning due to excessive intake of arsenic, which can lead to serious medical errors and legal problems. In recent years, there have been frequent reports of medical origin caused by arsenic-containing TCMs [5, 13, 18]. In many cases in China, criminal and civil convictions have been imposed based on the arsenic concentration of the drugs used and autopsy specimens [19]. In cases 1 and 2, the consumption or topical application of TCMs containing arsenic caused acute poisoning in the patients. In China, arsenic compounds were once widely used as pesticides in agriculture. The couple in cases 3 and 4 suffered acute arsenic poisoning from eating porridge dosed with arsenic-containing pesticides.

Gastrointestinal symptoms such as nausea, vomiting, abdominal pain, and dilute watery diarrhea are commonly

Fig. 1 Pathological features of acute arsenic poisoning in the liver, kidney, heart, and spleen **a** microvesicular steatosis in the peripheral areas of the hepatic lobules of case 3 (HE, 80 $\times$, 200 $\times$); **b** numerous small cytoplasmic vesicles of hepatocytes which were identified as lipid droplets in case 3 (ORO, 400 $\times$); **c** acute tubular necrosis and massive protein casts of case 4 (HE, 200 $\times$); **d** necrosis of the collecting duct in the renal inner medulla of case 4 (HE, 200 $\times$); **e** myocardial constriction band necrosis of case 3 (HE, 200 $\times$, 400 $\times$); **f** neutrophil infiltration (arrow) of the splenic corpuscles of case 2 (HE, 400 $\times$)



associated with acute arsenic poisoning. These symptoms are followed by the progression of liver and kidney dysfunction and myocardial injury [5, 10, 13, 14, 16], which is consistent with the symptoms of cases 1, 3, and 4 in this study. Other clinical features, such as numbness in the extremities [14] and diffuse skin rash [13, 16], have also been reported. In case 2, the prominent symptoms were localized swelling, bruising, and pain in the anal skin where the drug was topically applied. Vomiting occurred despite the absence of oral administration, suggesting that the gastrointestinal symptoms were not solely due to the direct corrosive effects of arsenic. The dermal absorption of As is low, only 1–6% [13], which allowed case 2 to have milder clinical symptoms with a longer survival time.

In fatal arsenic poisoning cases, the concentration of arsenic ranged from 0.16 to 41 $\mu\text{g/mL}$ in blood, 2.6 to 1635 $\mu\text{g/g}$ in the liver, and 1 to 1217 $\mu\text{g/g}$ in the kidney [5, 10, 13–16]. It is a common view that the liver and kidney

have the highest concentrations of arsenic due to their role in the metabolism and excretion of the substance. The highest concentrations of arsenic in the gastric wall and stomach contents have been observed for orally ingested inorganic arsenic [5, 15]. Additionally, the highest concentrations of arsenic have also been observed in the skin, through which arsenic is absorbed [13]. The data in Table 3 show that the highest concentration of arsenic was located in the liver, except at the site of direct exposure, suggesting that the liver is the primary target of arsenic. The concentration of arsenic in the kidney was three times that of the liver, as seen in an intravenous arsenic injection case [10]. Thus, the excretory function of the kidney may play a more dominant role in removing arsenic when it enters the bloodstream directly. The concentration of arsenic in the blood has always been an indicator of fatal arsenic cases. Blood arsenic concentrations drop rapidly because the average half-life of arsenic in the blood is 17–18 h [20]. Treatments such as continuous renal

Table 3 Arsenic concentration in postmortem specimens in these cases and literature

Postmortem specimens	Cardiac blood	Liver	Kidney	Stomach content	Lung	Brain	Heart	Spleen	Others
[5]	10.2	23.1	-	-	-	-	-	-	Gastric wall: 32.3 Pills: 372,000
[10]	0.629	38.250	127.000	1.319	2.780	-	-	-	Peripheral blood: 0.639 Urine: not detected Injection site: radial venous: 0.910 Injection site: cubital venous: 2.290
[13]	1.76	-	-	-	-	-	-	-	Back skin: 4.71 Chest skin: 4.23 Right leg skin: 5.68 Ointment: 60,100
[14]	0.525	39	-	-	-	-	-	-	Urine: 4.838 Hair: 74
[15]	3.00	1623	971	101,449	-	-	-	-	Duodenum content: 79,420
	20.1	1635	1217	23,369	-	-	-	-	Large intestine content: 19,564
	16.2	62.8	23.5	19,410	-	0.69	-	-	Duodenum content: 228,200
[16]	-	2.6	-	-	0.7	1.5	1.1	0.9	Peripheral blood: 0.355 Muscle: 1.9 Gastric wall: 0.9 Hair: 67 Nails: 13.8 Thyroid: 6.6 Pancreas: 0.6 Gall bladder: 0.5
Case 1*	0.95	14.63	12.90	-	1.61	1.07	1.58	3.27	Muscle: 1.35 Thymus: 0.72 Uncooked herbs: 9101
Case 2*	0.6	6	5	0.5	0.8	0.8	1	1	Bile: 1 Vitreous humor: 0.05 Peripheral blood: 0.2 Anal tissue: 100 Powder: 7000
Case 3*	1.1	9.9	-	2.7	-	-	-	-	Porridge: 4834
Case 4*	1.3	28	-	8	-	-	-	-	Porridge: 4834

Data are in µg/mL or µg/g

*: The cases in the present study

-: Not tested

replacement therapy can cause a further drop in blood arsenic concentration. Determination of blood arsenic concentration may not reflect the actual status of arsenic poisoning in such cases. Arsenic concentrations in the liver and kidneys tend to peak later and decline more slowly due to the kinetics of arsenic metabolism [21]. For this consideration, we suggest that the identification of arsenic poisoning cases can be improved by detecting arsenic concentrations in the liver and kidneys in addition to blood arsenic concentrations to increase reliability.

The strong affinity of As(III) for sulfhydryl groups is an essential factor in the acute toxicity of arsenic. By binding to sulfhydryl groups, As(III) inhibits the activity of several cellular enzymes, interferes with the citric acid cycle, and activates oxidative stress, which causes mitochondrial

damage [22–24]. As(V), on the other hand, causes further mitochondrial damage and inhibition of ATP production by replacing phosphate in oxidative phosphorylation [22–24]. Arsenic is toxic to most organ systems [24]. In the liver, severe mitochondrial damage can lead to the production of microvesicular steatosis [25, 26], consistent with the phenomena observed in cases 3 and 4. While hepatic steatosis is commonly described in previous case reports of arsenic poisoning, microvesicular steatosis in the peripheral areas of the hepatic lobules is less frequently depicted. Hepatocellular necrosis, cholestasis, and other pathological changes are associated with arsenic-induced oxidative stress [27]. High doses of arsenic intake lead to renal tubular necrosis and cause acute kidney injury [28]. The results of arsenic detection, pathological changes in the kidney, and clinical

laboratory tests were consistent with these findings. The detection of arsenic in the heart, abnormal myocardial enzyme profile, and pathological changes such as myocardial constriction band necrosis confirmed the myocardial toxicity of arsenic. In case 2, acute splenitis was observed, which has not been reported in other cases. A mouse study also showed that acute arsenic poisoning induces an inflammatory response in the spleen [29]. To our knowledge, acute arsenic toxicity in the spleen and the specific mechanisms require further investigation.

The toxicity of As is highly dependent on its solubility and valence. As(III), which has high solubility, is more toxic than As(V), which has low solubility [15]. Unfortunately, for various reasons, the valence and solubility of arsenic intoxicants and the valence and metabolite forms of arsenic in individual postmortem specimens were not analyzed in this study.

Conclusion

We report four cases of acute arsenic poisoning deaths and review the pathological changes and concentrations of arsenic in postmortem specimens in arsenic poisoning deaths over the past 20 years to provide a reference for identifying future arsenic poisoning deaths and to support further studies on arsenic toxicity. In TCM-related deaths, arsenic poisoning needs more attention, and arsenic in medicines should be more highly controlled and monitored. Hepatic microvesicular steatosis could be a significant indication of arsenic-related hepatotoxicity, and the impact of arsenic on the spleen may warrant further research. Finally, we propose that arsenic concentrations in the liver and kidneys may better complement those in the blood in identifying arsenic poisoning.

Key points

1. Four cases of fatal acute arsenic poisoning were reported.
2. Histopathological features of arsenic poisoning were summarized, and data on arsenic distribution were accumulated.
3. Arsenic concentrations in the liver and kidneys are of value in forensic practice.
4. TCM-related arsenic poisoning needs more attention in forensic practice.

Data availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Informed consent Written informed consent was obtained from the family members of the patient.

Conflict of interest The authors declare no competing interests.

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